

PARIKSHA365 — STUDY NOTES

# Physics — Common Book

SSC, RRB, State PSC, Banks (GA) — General  
Science

SSC

RRB

BANKS

STATE PSC

v 2026-04

**The promise.** Read this book end-to-end (with the usual two re-reads + active recall on the embedded prompts) and you will be able to attempt every quiz question in the Pariksha365 pool for this subject with a strong fundamental grasp.

# Examiner's Blueprint — How Physics Gets Tested

---

## ⚠ EXAM ALERT

Physics questions in competitive exams fall into a predictable pattern. They are rarely about solving equations — they are almost always about conceptual recall and application. The examiner's favourite moves are: \*\*1. "Why does X happen?" questions\*\* (pressure cooker, ice floating, sky colour) — these are pure understanding questions, and they appear in every paper. \*\*2. "Who discovered / invented X?"\*\* — Röntgen, Faraday, Chadwick, Fleming; these appear 1-2 times per paper. \*\*3. Correction for vision defects\*\* — myopia/hypermetropia and concave/convex lens. Never skipped by examiners. \*\*4. Motor vs Generator\*\* (which is which, which hand rule) — a classic 1-mark gift if you know it. \*\*5. Sound speed and echo distance\*\* — straightforward calculation questions. \*\*6. Alpha, Beta, Gamma radiation properties\*\* — penetration, charge, what stops them. Pure numerics (actual force calculations, wave equation solving) appear rarely. This book builds your conceptual understanding first, then gives you the numbers you must know cold.

## Priority order for your study time:

1. Daily-life phenomena (Part X) — "why" questions; one in every paper
  2. Instruments table (Part V) — 1-2 direct questions per paper
  3. Discoverers and inventions (Part W) — constant presence
  4. Sound: speed, echo, ranges (Part D) — easy marks
  5. Light: mirror/lens types, vision defects, Rayleigh scattering (Part D)
  6. Newton's laws with examples (Part A)
  7. Electricity: Ohm's law, Indian supply, AC/DC (Part E)
  8. Electromagnetism: motor/generator/transformer (Part E)
  9. Nuclear: fission/fusion/radiation types (Part F)
  10. Units, constants, formulas (Appendices)
-

# PART A — MECHANICS

## Chapter A1 — SI Units

The scientific world agreed on one system of measurement — the **International System of Units (SI)**. Here are its 7 base units, which all other units are built from.

| Quantity            | Unit     | Symbol |
|---------------------|----------|--------|
| Length              | metre    | m      |
| Mass                | kilogram | kg     |
| Time                | second   | s      |
| Electric current    | ampere   | A      |
| Temperature         | kelvin   | K      |
| Amount of substance | mole     | mol    |
| Luminous intensity  | candela  | cd     |

### © MNEMONIC

**\*\*"My Kind Sister Always Keeps Many Candles"\*\*** — Mass (kg), Kind (K = kelvin), Sister (s = second), Always (A = ampere), Keeps (this is a connector), Many (m = metre), Candles (cd = candela). Or more directly: **\*\*metre, kilogram, second, ampere, kelvin, mole, candela\*\*** — "Mr. K Sees Amphibians Killing Many Critters."

## Derived Units to Know

| Quantity        | Unit    | Symbol   | In base units                           |
|-----------------|---------|----------|-----------------------------------------|
| Force           | newton  | N        | $\text{kg} \cdot \text{m} / \text{s}^2$ |
| Pressure        | pascal  | Pa       | $\text{N} / \text{m}^2$                 |
| Energy / Work   | joule   | J        | $\text{N} \cdot \text{m}$               |
| Power           | watt    | W        | $\text{J} / \text{s}$                   |
| Frequency       | hertz   | Hz       | $\text{s}^{-1}$                         |
| Electric charge | coulomb | C        | $\text{A} \cdot \text{s}$               |
| Voltage         | volt    | V        | $\text{J} / \text{C}$                   |
| Resistance      | ohm     | $\Omega$ | $\text{V} / \text{A}$                   |

#### ✓ KEY FACT

The commercial unit of electrical energy is 1 kilowatt-hour (kWh) = 1 unit on your electricity bill =  $3.6 \times 10^6$  joules. 1 horsepower = 746 watts (not 750 — examiners use this to catch those who round up carelessly).

## Chapter A2 — Newton's Laws of Motion

Isaac Newton formulated these three laws in 1687 in his Principia Mathematica. They governed everything from cricket balls to rockets until Einstein refined them for extreme speeds and masses.

### First Law — Law of Inertia

An object at rest stays at rest, and an object moving at constant velocity continues moving at constant velocity, unless acted upon by an external force.

**Inertia** is the tendency of an object to resist change in its state of motion. Inertia is proportional to mass — a truck is harder to start moving than a bicycle.

#### ⚠ EXAM ALERT

Exam applications of Newton's First Law: seat belts (body wants to continue moving when car stops), pulling a tablecloth quickly (plates stay due to inertia), coins on a card flicked away (coins fall into the glass), dust flying off a beaten carpet. Any "lurching" or "jerking" scenario = First Law.

### Second Law — $F = ma$

The rate of change of momentum of an object is proportional to the applied force and in the direction of the force.

Simplified: **Force = mass × acceleration**

- More force → more acceleration.
- More mass → less acceleration for same force.
- SI unit of force: newton (N). One newton accelerates 1 kg at  $1 \text{ m/s}^2$ .

### Third Law — Action and Reaction

For every action there is an equal and opposite reaction. The two forces act on different objects.

### ⚠ EXAM ALERT

Third Law applications that appear in exams: **rocket propulsion** (gas pushed backward → rocket pushed forward), **gun recoil** (bullet pushed forward → gun pushes back), **swimming** (you push water backward → water pushes you forward), **helicopter tail rotor** (counters torque from main rotor). Jet engines, walking, and even rowing a boat are Third Law applications.

## Equations of Motion (Constant Acceleration)

- $v = u + at$
- $s = ut + \frac{1}{2}at^2$
- $v^2 = u^2 + 2as$

Where:  $u$  = initial velocity,  $v$  = final velocity,  $a$  = acceleration,  $t$  = time,  $s$  = displacement.

## Chapter A3 — Gravitation

### Newton's Universal Law

Every two masses attract each other with a force proportional to the product of their masses and inversely proportional to the square of the distance between them.

$$F = G \times (m_1 \times m_2) / r^2$$

$G = 6.674 \times 10^{-11} \text{ N}\cdot\text{m}^2/\text{kg}^2$  (Cavendish first measured this in 1798).

### Acceleration Due to Gravity ( $g$ )

- $g$  at Earth's surface =  $9.8 \text{ m/s}^2$  (use 10 for quick calculations).
- $g$  is **less at the equator** than at the poles — because: (1) the Earth is oblate (bulges at equator, so equatorial surface is farther from the centre), and (2) rotational effect reduces effective gravity.
- $g$  **decreases** as you go higher (away from centre).
- $g$  **decreases** as you go deeper (less mass pulling you).
- **At Earth's centre,  $g = 0$ .**

### ✓ KEY FACT

Escape velocity from Earth = **11.2 km/s** (~40,300 km/h). This is the minimum speed an object needs to permanently escape Earth's gravity without further propulsion. Orbital velocity at Earth's surface  $\approx 7.9 \text{ km/s}$  — the speed needed to maintain a circular orbit just above the surface.

## Kepler's Laws of Planetary Motion

1. **Law of Orbits** — Planets move in elliptical orbits with the Sun at one focus.
2. **Law of Areas** — A line joining a planet to the Sun sweeps out equal areas in equal time intervals (planets move faster when closer to the Sun).
3. **Law of Periods** —  $T^2 \propto r^3$  (square of orbital period is proportional to cube of semi-major axis).

## Satellites

- A **geostationary satellite** orbits at  $\sim 36,000$  km above the equator, completing one orbit in exactly 24 hours — so it appears stationary relative to Earth's surface. Used for telecommunications and weather.
  - **Weightlessness** in a satellite — the satellite and everything inside are in free fall together. The apparent weight is zero because both the person and the floor accelerate downward at the same rate.
- 

## Chapter A4 — Work, Energy, Power

- **Work** = Force  $\times$  displacement  $\times \cos(\theta)$ . Work is done only when displacement occurs in the direction of force. Unit: joule (J).
- **Kinetic Energy (KE)** =  $\frac{1}{2}mv^2$
- **Potential Energy (gravitational PE)** =  $mgh$
- **Power** = Work done / Time taken. Unit: watt (W).  $P = F \times v$ .

## Conservation of Energy

Energy cannot be created or destroyed. It transforms from one form to another. Total energy of an isolated system is constant.

Classic example: A pendulum at its highest point has maximum PE and zero KE. At the lowest point, PE converts entirely to KE. The total (PE + KE) stays constant throughout the swing (ignoring air resistance).

---

# PART B — PROPERTIES OF MATTER

---

## Chapter B1 — Pressure, Fluids, Buoyancy

- **Pressure** = Force / Area. SI unit: pascal (Pa). A sharp knife cuts better than a blunt one because the same force is applied over a smaller area.
- **Atmospheric pressure** at sea level = 101,325 Pa  $\approx$  1 atm = 760 mm Hg.
- Atmospheric pressure decreases with altitude (less air column above).

### Pascal's Law

Pressure applied to an enclosed fluid transmits equally in all directions. This is why hydraulic systems (car brakes, JCB excavators, dental chairs) can multiply force enormously with a small input.

### Archimedes' Principle

The upward buoyant force on a body immersed in fluid equals the weight of fluid displaced by the body.

A steel ship floats because it is hollow — its effective density (steel + air) is less than water, so it displaces enough water for the buoyant force to equal its weight. A solid steel ball sinks because its density is much greater than water.

#### EXAM ALERT

Ice floats on water — this is anomalous expansion. Water expands on freezing (unusual behaviour). Ice density is 0.92 g/cm<sup>3</sup> vs water at 1.0 g/cm<sup>3</sup>. This is why fish survive winter — ice floats and forms an insulating layer; liquid water at 4 °C stays below, where fish live.

## Chapter B2 — Surface Tension, Viscosity, Capillarity

- **Surface tension** — why water forms droplets (spherical = minimum surface area for a given volume), why insects can walk on water, why a needle can float on still water. Detergent reduces surface tension — the soap molecule disrupts the surface layer.
  - **Viscosity** — internal friction in fluids. Honey > water > petrol. Viscosity decreases with temperature (why engine oil thins when hot).
  - **Capillarity** — rise or fall of liquid in narrow tubes due to surface tension vs gravity. Water rises in a glass capillary (adhesion to glass > cohesion between water molecules). Mercury falls in a glass capillary. Plants use capillary action to draw water up roots and stems.
-

# PART C — HEAT AND THERMODYNAMICS

## Chapter C1 — Temperature Scales

| Scale           | Freezing of water | Boiling of water |
|-----------------|-------------------|------------------|
| Celsius (°C)    | 0                 | 100              |
| Fahrenheit (°F) | 32                | 212              |
| Kelvin (K)      | 273.15            | 373.15           |

Conversion:  $^{\circ}\text{F} = (9/5 \times ^{\circ}\text{C}) + 32$ . Kelvin = Celsius + 273.15.

Absolute zero (0 K =  $-273.15^{\circ}\text{C}$ ) — theoretically the coldest possible temperature; particles have minimum possible energy.

## Chapter C2 — Heat Transfer

| Mode       | Medium needed      | Mechanism                         | Examples                                          |
|------------|--------------------|-----------------------------------|---------------------------------------------------|
| Conduction | Solid (mainly)     | Particle vibration, electron flow | Metal rod heating, iron; ice cube melting in hand |
| Convection | Fluid (liquid/gas) | Bulk movement of fluid            | Boiling water, sea breeze, room air conditioning  |
| Radiation  | None — vacuum OK   | Electromagnetic waves             | Sun warming Earth, warmth from a fire             |

### ✓ KEY FACT

The Sun's heat reaches Earth through the vacuum of space — this is radiation (electromagnetic waves). Only radiation requires no medium. This is why a thermos (vacuum flask) has a silvered interior — it prevents heat loss by all three mechanisms: vacuum stops conduction and convection, silvering reflects radiation.

## Chapter C3 — Specific Heat and Latent Heat

- **Specific heat** — heat energy needed to raise 1 kg of a substance by  $1^{\circ}\text{C}$  (or 1 K).
- Water:  $4184 \text{ J/kg}\cdot^{\circ}\text{C}$  — the highest among common substances. This is why coastal climates are milder — oceans absorb and release enormous heat with little temperature change.
- Iron:  $\sim 449 \text{ J/kg}\cdot^{\circ}\text{C}$ . Aluminium:  $\sim 900 \text{ J/kg}\cdot^{\circ}\text{C}$ .



- **Latent heat** — heat absorbed or released during a phase change (state change) WITHOUT any temperature change.
- Latent heat of fusion of ice: 334 J/g (ice → water at 0 °C).
- Latent heat of vaporisation of water: 2260 J/g (water → steam at 100 °C).

#### ⚠ EXAM ALERT

Steam burns are worse than boiling water burns at the same 100 °C — because steam releases its latent heat of vaporisation (2260 J/g) when it condenses on your skin, in addition to the heat from cooling. We sweat to cool down because sweat evaporating absorbs latent heat of vaporisation from the skin surface.

## Chapter C4 — Laws of Thermodynamics

1. **Zeroth Law** — If system A is in thermal equilibrium with system B, and B with C, then A and C are also in equilibrium. This defines temperature.
2. **First Law** —  $\Delta U = Q - W$ . Energy is conserved. Heat added to a system equals change in internal energy plus work done by the system.
3. **Second Law** — Heat cannot spontaneously flow from cold to hot. Entropy (disorder) of an isolated system always increases.
4. **Third Law** — At absolute zero, entropy of a perfect crystal is zero. Absolute zero is unreachable in practice.

A refrigerator moves heat from cold to hot — but only by consuming electricity (external work). This is the Second Law in action.

---

# PART D — WAVES, SOUND, AND LIGHT

## Chapter D1 — Sound

Sound is a mechanical wave — it requires a medium to travel. It cannot travel through vacuum. That is why space is silent.

### Speed of Sound

| Medium       | Speed (approximate) |
|--------------|---------------------|
| Air at 0 °C  | 331 m/s             |
| Air at 20 °C | 343 m/s             |
| Water        | ~1500 m/s           |
| Steel        | ~5000 m/s           |

Sound travels faster in solids than in liquids, and faster in liquids than in gases. Speed increases with temperature.

### Frequency Ranges

| Range           | Frequency          | Who uses it                                     |
|-----------------|--------------------|-------------------------------------------------|
| Infrasonic      | Below 20 Hz        | Whales, elephants, earthquakes                  |
| Audible (human) | 20 Hz to 20,000 Hz | Humans                                          |
| Ultrasonic      | Above 20,000 Hz    | Bats, dolphins, dogs; medical sonography, SONAR |

#### ✓ KEY FACT

Echo needs a minimum distance of **\*\*17 metres\*\*** between you and the reflecting surface. Why? Sound takes 0.1 seconds to reach the reflector and return (minimum time for the human ear to distinguish echo from original). At 343 m/s, that is  $343 \times 0.1 / 2 = 17.15$  m.

### Doppler Effect

The apparent frequency of a sound changes when the source and observer are in relative motion. When the source approaches, frequency appears higher (pitch rises). When the source moves away, frequency appears lower (pitch falls). Classic example: an ambulance siren changes pitch as it passes you.

### SONAR

Sound Navigation and Ranging — uses ultrasonic pulses to detect underwater objects and map the ocean floor. The time for the echo to return, multiplied by the speed of sound in water,

gives twice the distance to the object.

---

## Chapter D2 — Light and Optics

Light is an electromagnetic wave that travels at  $c = 3 \times 10^8 \text{ m/s}$  in vacuum. This is the fastest speed in the universe.

Speed of light in other media: water  $\sim 2.25 \times 10^8 \text{ m/s}$ , glass  $\sim 2 \times 10^8 \text{ m/s}$ . Light slows down in denser media.

**Refractive index ( $n$ )** = speed of light in vacuum / speed of light in medium. For glass  $\approx 1.5$ , water  $\approx 1.33$ , diamond  $\approx 2.42$ .

### Reflection

- Angle of incidence = angle of reflection.
- Incident ray, reflected ray, and normal are in the same plane.

### Refraction (Snell's Law)

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

Light bends toward the normal when entering a denser medium and away from the normal when entering a less-dense medium.

**Total Internal Reflection (TIR)** — when light travels from a denser to a less-dense medium at an angle greater than the critical angle, it is completely reflected back. This is what gives diamonds their sparkle (very small critical angle  $\sim 24^\circ$ ) and what optical fibres rely on.

### Mirrors — Which Type Does What

| Mirror               | Image formed                                              | Used in                                                             |
|----------------------|-----------------------------------------------------------|---------------------------------------------------------------------|
| Plane                | Same size, virtual, erect, laterally inverted             | Looking glass                                                       |
| Concave (converging) | Depends on object position; real+inverted if beyond focus | Shaving mirror, headlights, dentist's mirror, reflecting telescopes |
| Convex (diverging)   | Always virtual, erect, diminished                         | Rear-view mirrors in vehicles, security mirrors                     |

#### ⚠ EXAM ALERT

Rear-view mirrors in vehicles are always convex — they give a wider field of view (the image is diminished but you see more). The "objects in mirror are closer than they appear" warning is because the convex mirror makes distant objects appear farther away. A driver checking a concave rear-view mirror would get a misleading sense of distances.

## Lenses — Which Type Does What

| Lens                | Image type                            | Used for                                                           |
|---------------------|---------------------------------------|--------------------------------------------------------------------|
| Convex (converging) | Real or virtual depending on position | Magnifying glass, cameras, projectors, microscopes, eye's own lens |
| Concave (diverging) | Always virtual, erect, diminished     | Correcting myopia (short-sightedness)                              |

Lens formula:  $1/v - 1/u = 1/f$ . Power of lens:  $P = 1/f$  (metres). Unit: dioptre (D).

## Defects of Vision and Corrections

| Defect                           | What happens                                             | Correction               |
|----------------------------------|----------------------------------------------------------|--------------------------|
| Myopia (short-sightedness)       | Image forms in front of retina; can see near but not far | Concave (diverging) lens |
| Hypermetropia (long-sightedness) | Image forms behind retina; can see far but not near      | Convex (converging) lens |
| Presbyopia                       | Eye lens stiffens with age; cannot adjust                | Bifocal lens             |
| Astigmatism                      | Cornea unevenly curved                                   | Cylindrical lens         |
| Cataract                         | Lens becomes opaque                                      | Surgery to replace lens  |

### ⦿ MNEMONIC

**\*\*Myopia → Concave, Hypermetropia → Convex\*\***. Memory trick: the words **\*\*Myopia\*\*** and **\*\*Concave\*\*** both have fewer common letters to remember — just link the **\*\*M\*\*yopic** person needs a **\*\*C\*\*oncave** lens. Or: Myopia = "My can't see far" → need to diverge the light more → concave.

## Dispersion and Colour

- **Dispersion** — white light splits into its spectrum (VIBGYOR) when passing through a prism. Newton first demonstrated this in 1666.
- **VIBGYOR**: Violet, Indigo, Blue, Green, Yellow, Orange, Red — in order of decreasing wavelength (increasing frequency). Violet has highest frequency in visible range.
- **Rainbow** — caused by refraction + total internal reflection + dispersion inside raindrops.
- **Blue sky** — Rayleigh scattering. Shorter (blue) wavelengths scatter  $\sim 5\times$  more than longer (red) wavelengths. So when sunlight enters the atmosphere, blue light scatters in all directions, making the whole sky look blue.
- **Red sunset** — at sunrise/sunset, sunlight travels through much more atmosphere to reach your eyes. Blue light has already scattered away; red, which scatters least, dominates.

#### ✓ KEY FACT

Diamond sparkles because its refractive index is very high (2.42), giving a very small critical angle ( $\sim 24^\circ$ ). Most light rays entering the diamond hit a face at more than  $24^\circ$  and undergo total internal reflection repeatedly — they bounce inside the diamond and finally exit from the top. This internal bouncing creates the famous sparkle.

## Electromagnetic Spectrum

From lowest to highest frequency (highest to lowest wavelength):

Radio → Microwaves → Infrared → Visible light → Ultraviolet → X-rays → Gamma rays

| Band             | Wavelength    | Key use                                            |
|------------------|---------------|----------------------------------------------------|
| Radio            | km to m       | AM/FM radio, TV broadcasting                       |
| Microwave        | cm to mm      | Microwave ovens, mobile phones, radar              |
| Infrared (IR)    | $\mu\text{m}$ | TV remotes, thermal cameras, heat sensing          |
| Visible          | 400–700 nm    | Human vision (VIBGYOR)                             |
| Ultraviolet (UV) | <400 nm       | Sterilisation, vitamin D synthesis, fluorescence   |
| X-rays           | 0.01–10 nm    | Medical imaging, airport security                  |
| Gamma            | <0.01 nm      | Cancer radiotherapy, sterilising medical equipment |

# PART E — ELECTRICITY AND MAGNETISM

---

## Chapter E1 — Current Electricity

- **Charge (q)** — unit: coulomb (C).
- **Current (I)** = charge per second =  $q/t$ . Unit: ampere (A).
- **Voltage / Potential difference (V)** — the "electrical pressure" that drives current. Unit: volt (V).
- **Resistance (R)** — opposition to current. Unit: ohm ( $\Omega$ ).

### Ohm's Law

$$V = I \times R$$

At constant temperature, current through a conductor is proportional to the voltage across it.

Power consumed:  $P = V \times I = I^2R = V^2/R$ . Unit: watt (W).

### Household Electricity in India

**230 V, 50 Hz, AC.** The 230 V is the RMS voltage. USA uses 120 V, 60 Hz.

- **AC (Alternating Current)** — current reverses direction periodically (50 times per second in India = 50 Hz). Used for transmission over long distances because voltage can be stepped up or down using transformers.
- **DC (Direct Current)** — current flows in one direction only. Used in batteries, electronic devices.

#### ⚠ EXAM ALERT

Why is AC preferred for power transmission? Transformers can step up voltage for long-distance transmission (reducing current  $\rightarrow$  reducing heat loss in wires:  $P = I^2R$ ). At the destination, step it back down for safe home use. Transformers work only on AC, not DC — because DC doesn't create a changing magnetic flux, so no induction occurs.

### Circuits and Safety

- **Fuse** — a thin wire of low-melting-point alloy that melts when current exceeds a safe limit, breaking the circuit.
- **MCB (Miniature Circuit Breaker)** — modern alternative; trips a switch instead of burning a wire; can be reset.
- **Short circuit** — accidental path of near-zero resistance; causes very high current  $\rightarrow$  heat  $\rightarrow$  fire.
- **Earthing** — safety connection to the ground. Any leakage current flows harmlessly into the ground instead of through a person.

# Chapter E2 — Magnetism and Electromagnetic Induction

## Magnets

- A bar magnet has two poles — North and South. Like poles repel; unlike poles attract.
- Earth is a giant magnet. The geographic North Pole of Earth is near the magnetic South Pole (that's why the North-seeking pole of a compass needle points north — it's attracted to Earth's magnetic south).
- Magnetic field lines run from North to South outside the magnet (and South to North inside).

## Electromagnetic Induction — Faraday's Discovery (1831)

Moving a magnet near a coil of wire induces a voltage (EMF) in the coil, and therefore a current if the circuit is complete.

✓ KEY FACT

This is the foundation of everything electrical we use. Generators in power stations, microphones, guitar pickups, wireless charging — all are applications of electromagnetic induction. Michael Faraday made this discovery in 1831, and it changed civilisation.

## Lenz's Law

The direction of the induced current is always such that it opposes the change (motion or flux change) causing it. Think of it as electromagnetic braking — nature resists change.

## Motor and Generator

| Device             | Converts                | Principle                                                          | Hand Rule                 |
|--------------------|-------------------------|--------------------------------------------------------------------|---------------------------|
| Electric motor     | Electrical → Mechanical | Current-carrying conductor in a magnetic field experiences a force | Fleming's Left-Hand Rule  |
| Generator / Dynamo | Mechanical → Electrical | Moving conductor in a magnetic field induces current               | Fleming's Right-Hand Rule |

🕒 MNEMONIC

**\*\*Fleming's Left-Hand Rule for Motor\*\***: Hold your left hand with thumb, index finger, and middle finger all at right angles to each other. Index finger = direction of magnetic Field, Middle finger = direction of Current, Thumb = direction of Thrust (motion/force on conductor). For a generator, use the Right hand with the same logic — Motion thumb, Field index, Current middle. Motor = Left = I drive the **\*\*L\*\***eft side. Generator = Right.

## Transformer

Changes the voltage of AC supply.

- **Step-up transformer:** increases voltage, decreases current.
- **Step-down transformer:** decreases voltage, increases current.

Works on mutual induction — the changing current in the primary coil creates a changing magnetic field that induces a current in the secondary coil. Transformers work only with AC.

---



# PART F — MODERN PHYSICS

## Chapter F1 — Atomic Models

| Model              | Year | Proposer                 | Key idea                                                  |
|--------------------|------|--------------------------|-----------------------------------------------------------|
| Plum-pudding       | 1897 | J.J. Thomson             | Atom = positive sphere with electrons embedded like plums |
| Nuclear            | 1911 | Ernest Rutherford        | Most of atom is empty space; tiny dense nucleus at centre |
| Bohr model         | 1913 | Niels Bohr               | Electrons orbit in fixed quantised energy levels          |
| Quantum mechanical | 1926 | Schrödinger / Heisenberg | Electrons exist as probability clouds (orbitals)          |

Rutherford used the famous gold-foil (alpha-scattering) experiment — firing alpha particles at gold foil. Most passed through (empty space), some deflected slightly (electrons), and a few bounced straight back (nucleus). He concluded the atom is mostly empty with a tiny, dense, positive nucleus.

## Chapter F2 — Subatomic Particles

| Particle | Charge | Relative mass          | Discoverer     | Year |
|----------|--------|------------------------|----------------|------|
| Electron | -1     | 1/1837                 | J.J. Thomson   | 1897 |
| Proton   | +1     | 1                      | Rutherford     | 1919 |
| Neutron  | 0      | ~1 (slightly > proton) | James Chadwick | 1932 |

### ✓ KEY FACT

The heaviest sub-atomic particle is the **neutron** (just slightly heavier than the proton). Three quick-recall facts: electron → Thomson 1897, proton → Rutherford 1919, neutron → Chadwick 1932. The first Nobel Prize in Physics (1901) went to Röntgen for X-rays.

## Chapter F3 — Radioactivity

Henri Becquerel discovered radioactivity in 1896 (from uranium). Marie and Pierre Curie then isolated polonium (named after Marie's homeland Poland) and radium in 1898. Marie Curie won two Nobel Prizes — Physics 1903 and Chemistry 1911.

## Three Types of Radiation

| Type  | Symbol   | Nature                       | Charge | Penetrating power | Stopped by           |
|-------|----------|------------------------------|--------|-------------------|----------------------|
| Alpha | $\alpha$ | Helium nucleus (2p + 2n)     | +2     | Lowest            | Paper, few cm of air |
| Beta  | $\beta$  | Fast electron                | -1     | Medium            | 3mm aluminium        |
| Gamma | $\gamma$ | High-energy EM wave (photon) | 0      | Highest           | Several cm of lead   |

### ☉ MNEMONIC

Penetrating power order:  **$\alpha < \beta < \gamma$**  (alphabetical = weakest to strongest). What stops them: **Paper, Aluminium, Lead** — think PAL. Alpha needs a PAL to stop it (paper stops alpha). Aluminium stops beta. Lead stops gamma.

## Chapter F4 — Nuclear Reactions

### Fission

A heavy nucleus (Uranium-235 or Plutonium-239) splits into smaller fragments + 2-3 neutrons + enormous energy.

Each released neutron can trigger more fissions → chain reaction. In a nuclear reactor, control rods (cadmium or boron) absorb excess neutrons to control the rate. In an atomic bomb, the chain reaction is uncontrolled.

### Fusion

Light nuclei (hydrogen isotopes — deuterium and tritium) combine to form helium + energy.

This is the source of the Sun's energy. Fusion releases far more energy per unit mass than fission, but controlled fusion on Earth has not yet been achieved commercially. The ITER project in France aims to demonstrate controlled fusion.

### ⚠ EXAM ALERT

**"Sun's energy comes from nuclear fusion"** — this appears in almost every exam. Hydrogen (specifically deuterium and tritium) fuses to form helium, releasing energy according to Einstein's  $E = mc^2$ . The mass of the helium product is slightly less than the combined mass of the inputs — that tiny mass difference becomes an enormous amount of energy.

$$E = mc^2$$

Einstein's mass-energy equivalence. Even a tiny mass ( $m$ ) holds an enormous amount of energy ( $c = 3 \times 10^8$  m/s, so  $c^2 = 9 \times 10^{16}$ ). Burning 1 gram of matter completely would yield  $9 \times 10^{13}$  joules — equivalent to 20,000 tons of TNT.

## Radiocarbon Dating

Carbon-14 ( $^{14}\text{C}$ ) is a radioactive isotope of carbon with half-life of **5730 years**. Living organisms maintain a constant  $^{14}\text{C}/^{12}\text{C}$  ratio by continuously taking in carbon. When they die,  $^{14}\text{C}$  decays without being replenished. Measuring the remaining  $^{14}\text{C}$  ratio tells us the age of the organism. Used in archaeology (up to  $\sim 50,000$  years back).

## Chapter F5 — Semiconductors

| Type          | Resistivity  | Band gap             | Examples                     |
|---------------|--------------|----------------------|------------------------------|
| Conductor     | Very low     | None                 | Copper, silver, aluminium    |
| Semiconductor | Intermediate | Small ( $\sim 1$ eV) | Silicon (Si), Germanium (Ge) |
| Insulator     | Very high    | Large                | Glass, rubber, plastic       |

- **N-type semiconductor** — doped with pentavalent atoms (phosphorus, arsenic); extra electrons are charge carriers.
  - **P-type semiconductor** — doped with trivalent atoms (boron, aluminium); "holes" are charge carriers.
  - **Diode** (P-N junction) — allows current in one direction only.
  - **LED (Light Emitting Diode)** — emits light when forward current flows.
  - **Transistor** — invented 1947 by Bardeen, Brattain, Shockley (Bell Labs); Nobel 1956. Functions as amplifier and switch. Foundation of all modern electronics.
  - **Integrated Circuit (IC)** — many transistors on one silicon chip. Invented by Jack Kilby, 1958; Nobel 2000.
-

## PART V — INSTRUMENTS AND WHAT THEY MEASURE

---

| Instrument                | Measures                                               |
|---------------------------|--------------------------------------------------------|
| Ammeter                   | Electric current (connected in series)                 |
| Voltmeter                 | Potential difference / voltage (connected in parallel) |
| Galvanometer              | Very small electric currents                           |
| Multimeter                | Voltage, current, and resistance                       |
| Wattmeter                 | Electric power                                         |
| Energy meter              | Electrical energy consumed (kWh)                       |
| Ohmmeter                  | Electrical resistance                                  |
| Thermometer               | Temperature                                            |
| Pyrometer                 | Very high temperatures (furnaces, molten metal)        |
| Calorimeter               | Heat exchanged in reactions                            |
| Barometer                 | Atmospheric pressure                                   |
| Manometer                 | Gas pressure                                           |
| Hygrometer / Psychrometer | Relative humidity                                      |
| Anemometer                | Wind speed                                             |
| Rain gauge (Pluviometer)  | Rainfall                                               |
| Seismograph               | Earthquake intensity and location                      |
| Lactometer                | Purity / density of milk                               |
| Hydrometer                | Density of liquids                                     |
| Saccharometer             | Sugar content of a solution                            |
| Salinometer               | Salt concentration in water                            |
| Speedometer               | Speed of a vehicle                                     |
| Odometer                  | Distance travelled by a vehicle                        |
| Tachometer                | Rotational speed (RPM) of a shaft or engine            |
| Sextant                   | Angular altitude of celestial bodies (navigation)      |
| Theodolite                | Horizontal and vertical angles (land surveying)        |
| Audiometer                | Hearing sensitivity (threshold of hearing)             |

| Instrument                  | Measures                                          |
|-----------------------------|---------------------------------------------------|
| Stethoscope                 | Heart and lung sounds                             |
| Sphygmomanometer            | Blood pressure                                    |
| Cardiograph (ECG)           | Electrical activity of the heart                  |
| Electroencephalograph (EEG) | Electrical activity of the brain                  |
| Tonometer                   | Eye pressure (glaucoma screening)                 |
| Endoscope                   | Images from inside the body                       |
| Periscope                   | Viewing over obstacles (used in submarines)       |
| Spectrometer                | Wavelengths of light / analysis of light spectrum |
| Refractometer               | Refractive index of a substance                   |
| Photometer                  | Intensity of light                                |
| Bolometer                   | Power of incident electromagnetic radiation       |
| Oscilloscope                | Waveform of electrical signals                    |

---

## PART W — DISCOVERIES AND INVENTIONS IN PHYSICS

| Discovery / Invention                              | Scientist               | Year  | Country          |
|----------------------------------------------------|-------------------------|-------|------------------|
| Compound microscope                                | Zacharias Janssen       | 1590  | Netherlands      |
| Telescope (refractor)                              | Galileo Galilei         | 1609  | Italy            |
| Reflecting telescope                               | Isaac Newton            | 1668  | England          |
| Laws of motion, universal gravitation              | Isaac Newton            | 1687  | England          |
| Electrostatic machine (Wimshurst)                  | Robert Van de Graaff    | 1929  | USA              |
| Barometer                                          | Evangelista Torricelli  | 1643  | Italy            |
| Steam engine (improved)                            | James Watt              | 1769  | Scotland         |
| Battery (voltaic pile)                             | Alessandro Volta        | 1800  | Italy            |
| Electromagnetic induction                          | Michael Faraday         | 1831  | England          |
| Laws of thermodynamics (conservation of energy)    | Helmholtz, Joule, Mayer | 1840s | Germany, England |
| Telephone                                          | Alexander Graham Bell   | 1876  | Scotland/USA     |
| Electric light bulb (incandescent)                 | Thomas Edison           | 1879  | USA              |
| Wireless (radio) transmission                      | Guglielmo Marconi       | 1895  | Italy            |
| Wireless waves (millimetre waves)                  | Jagadish Chandra Bose   | 1895  | India            |
| X-rays                                             | Wilhelm Röntgen         | 1895  | Germany          |
| Radioactivity                                      | Henri Becquerel         | 1896  | France           |
| Polonium + Radium                                  | Marie and Pierre Curie  | 1898  | Poland/France    |
| Quantum hypothesis                                 | Max Planck              | 1900  | Germany          |
| Aeroplane (first flight)                           | Wright Brothers         | 1903  | USA              |
| Special Theory of Relativity, photoelectric effect | Albert Einstein         | 1905  | Germany          |
| Bohr atomic model                                  | Niels Bohr              | 1913  | Denmark          |
| Television (mechanical)                            | John Logie Baird        | 1925  | Scotland         |
| Raman Effect                                       | C.V. Raman              | 1928  | India            |
| Cyclotron (particle accelerator)                   | Ernest Lawrence         | 1929  | USA              |
| Nuclear chain reaction                             | Enrico Fermi            | 1942  | Italy/USA        |

| Discovery / Invention               | Scientist                         | Year      | Country       |
|-------------------------------------|-----------------------------------|-----------|---------------|
| Transistor                          | Bardeen, Brattain, Shockley       | 1947      | USA           |
| Holography                          | Dennis Gabor                      | 1948      | Hungary/UK    |
| Laser                               | Theodore Maiman                   | 1960      | USA           |
| Integrated circuit                  | Jack Kilby                        | 1958      | USA           |
| Optical fibre communication         | Charles Kao                       | 1966      | China/UK      |
| Gravitational waves detected (LIGO) | LIGO team (Thorne, Weiss, Barish) | 2015/2017 | USA           |
| Higgs boson confirmed               | CERN experiments                  | 2012      | International |
| Bose-Einstein statistics            | S.N. Bose + Einstein              | 1924      | India/Germany |
| Chandrasekhar limit                 | S. Chandrasekhar                  | 1930      | India/USA     |

#### ⚠ EXAM ALERT

Indian scientists to remember for physics: **C.V. Raman** (Raman Effect, Nobel Prize in Physics 1930 — the only Nobel in Physics won by an Indian as of 2026), **Jagadish Chandra Bose** (millimetre/microwave waves, 1895 — before Marconi in some interpretations), **Satyendra Nath Bose** (Bose-Einstein statistics, 1924 — the "boson" in Higgs boson is named after him), **Subrahmanyan Chandrasekhar** (Chandrasekhar limit, Nobel 1983).

## PART X — DAILY-LIFE PHYSICS ("WHY" QUESTIONS)

---

Examiners love this section. One to two questions on everyday physics phenomena appear in almost every competitive exam. Learn the explanation for each.

| Phenomenon                        | Explanation                                                                                                                                                         |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Pressure cooker cooks faster      | Higher pressure raises the boiling point of water to $\sim 120^\circ\text{C}$ ; food cooks faster at higher temperature                                             |
| Cooking takes longer on mountains | Atmospheric pressure is lower at high altitude; water boils at lower temperature ( $\sim 70^\circ\text{C}$ on Everest); food cooks slower                           |
| Ice floats on water               | Anomalous expansion — water expands on freezing; ice density ( $0.92\text{ g/cm}^3$ ) < water density ( $1.0\text{ g/cm}^3$ )                                       |
| Fish survive under frozen lake    | Ice forms on top and insulates the water below; liquid water at $\sim 4^\circ\text{C}$ stays at the bottom where fish live                                          |
| Mirage in desert                  | Light bends due to temperature gradient in hot air near the ground — total internal reflection of sky light                                                         |
| Stars twinkle, planets don't      | Stars are point sources — atmospheric turbulence causes their single ray to flicker. Planets are extended discs — their many rays average out, no twinkling         |
| Sky is blue                       | Rayleigh scattering — shorter (blue) wavelengths scatter much more than longer (red) wavelengths                                                                    |
| Sunset and sunrise are red/orange | Sunlight travels a long path through the atmosphere; blue has already scattered away; red, which scatters least, survives to reach your eyes                        |
| Rainbow forms after rain          | Water droplets act as tiny prisms — refraction + total internal reflection + dispersion separate white light into VIBGYOR                                           |
| Diamond sparkles                  | Very high refractive index (2.42) $\rightarrow$ small critical angle ( $\sim 24^\circ$ ) $\rightarrow$ total internal reflection of most rays $\rightarrow$ sparkle |
| Soap film shows colours           | Thin-film interference — light reflects from both surfaces of the thin film; they interfere constructively/destructively at different wavelengths                   |
| Why we sweat to cool              | Evaporation of sweat absorbs latent heat of vaporisation from the skin surface                                                                                      |
| Steam burn worse than water burn  | Steam at $100^\circ\text{C}$ releases latent heat of vaporisation ( $2260\text{ J/g}$ ) when it condenses on skin, plus heat from cooling                           |
| We slip on ice                    | Pressure of body weight on ice slightly lowers its melting point; a thin film of water forms $\rightarrow$ low friction                                             |



| Phenomenon                                    | Explanation                                                                                                                         |
|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Spoon appears bent in water                   | Refraction — light bends when crossing from water to air; the image appears to shift                                                |
| Lightning seen before thunder                 | Light travels at $3 \times 10^8$ m/s, sound at $\sim 340$ m/s; light arrives almost instantly, sound takes $\sim 3$ seconds per km  |
| Ceiling fan cools us                          | Increases evaporation of sweat from skin; does not actually cool the room air                                                       |
| Hot tea in saucer cools faster                | Larger surface area increases rate of evaporation and radiation                                                                     |
| Pendulum clock loses time in summer           | Pendulum rod expands with heat $\rightarrow$ longer pendulum $\rightarrow$ longer period $\rightarrow$ clock runs slow              |
| Birds sit on high-tension wires safely        | Both feet are on the same wire at the same potential; no potential difference between them $\rightarrow$ no current through bird    |
| A 3-pin plug is safer                         | Earth pin diverts any leakage current to the ground instead of through the user                                                     |
| Microwave heats food                          | Microwaves make water molecules vibrate (dielectric heating); heating from inside out                                               |
| Helicopter needs a tail rotor                 | Main rotor rotation creates torque on the body (Newton's 3rd Law); tail rotor counters this torque                                  |
| Geostationary satellite stays above same spot | Orbital period is 24 hours, same as Earth's rotation; appears stationary from Earth                                                 |
| Satellites feel weightless                    | Satellite and astronaut are in free fall together; apparent weight = 0 (no contact force needed)                                    |
| Echo minimum distance is 17 m                 | Human ear needs 0.1 s gap to distinguish echo from original sound; at 340 m/s, minimum round trip = 34 m $\rightarrow$ 17 m one way |
| Mountaineers carry oxygen cylinders           | Atmospheric pressure at high altitude is very low; partial pressure of oxygen is insufficient for breathing                         |

# PART AA — KEY CONSTANTS AND FORMULAS

---

## Physical Constants

| Constant                                    | Symbol   | Value                                                        |
|---------------------------------------------|----------|--------------------------------------------------------------|
| Speed of light in vacuum                    | c        | $3 \times 10^8 \text{ m/s}$                                  |
| Acceleration due to gravity (Earth surface) | g        | $9.8 \text{ m/s}^2$                                          |
| Universal gravitational constant            | G        | $6.674 \times 10^{-11} \text{ N}\cdot\text{m}^2/\text{kg}^2$ |
| Planck's constant                           | h        | $6.626 \times 10^{-34} \text{ J}\cdot\text{s}$               |
| Charge of electron                          | e        | $1.6 \times 10^{-19} \text{ C}$                              |
| Mass of electron                            | $m_e$    | $9.11 \times 10^{-31} \text{ kg}$                            |
| Mass of proton                              | $m_p$    | $1.67 \times 10^{-27} \text{ kg}$                            |
| Avogadro's number                           | $N_a$    | $6.022 \times 10^{23}$                                       |
| Gas constant                                | R        | $8.314 \text{ J/mol/K}$                                      |
| Boltzmann constant                          | k        | $1.38 \times 10^{-23} \text{ J/K}$                           |
| Stefan-Boltzmann constant                   | $\sigma$ | $5.67 \times 10^{-8} \text{ W/m}^2/\text{K}^4$               |
| Atmospheric pressure                        | —        | $101,325 \text{ Pa} \approx 760 \text{ mm Hg}$               |

## Key Non-SI Conversions

- 1 horsepower (HP) = 746 watts
- 1 kWh (1 unit of electricity) =  $3.6 \times 10^6$  joules
- 1 calorie = 4.184 joules
- 1 eV (electron volt) =  $1.6 \times 10^{-19} \text{ J}$
- 1 light-year =  $9.46 \times 10^{15}$  metres
- 1 ångström =  $10^{-10}$  metres
- $0^\circ\text{C} = 273.15 \text{ K} = 32^\circ\text{F}$

## Important Numerical Values to Memorise

- Escape velocity from Earth = **11.2 km/s**
- Orbital velocity at Earth surface = **7.9 km/s**
- Geostationary orbit altitude = **36,000 km**
- Speed of sound in air ( $20^\circ\text{C}$ ) = **343 m/s**
- Specific heat of water = **4184 J/kg/°C**
- Latent heat of fusion of ice = **334 J/g**
- Latent heat of vaporisation of water = **2260 J/g**
- India household supply = **230 V, 50 Hz AC**

- Refractive index of diamond = **2.42**
- Refractive index of water = **1.33**

## Formula Quick Reference

**Mechanics:**  $F = ma$  |  $W = mgh$  |  $KE = \frac{1}{2}mv^2$  |  $P = W/t$  |  $v^2 = u^2 + 2as$

**Heat:**  $Q = mc\Delta T$  |  $Q = mL$  |  $F = (9/5)C + 32$  |  $K = C + 273$

**Light:**  $n = c/v_{\text{medium}}$  |  $\sin(\theta_c) = 1/n$  |  $1/v - 1/u = 1/f$  |  $P = 1/f$  (diopetre)

**Electricity:**  $V = IR$  |  $P = VI = I^2R$  |  $E = Pt$  | series:  $R = R_1 + R_2$  | parallel:  $1/R = 1/R_1 + 1/R_2$

**Modern:**  $E = mc^2$  |  $E = hf$  |  $\lambda = h/p$  (de Broglie)

---

## APPENDIX A — 25-Question Mini-Mock

---

Attempt all 25 without looking at answers. Time yourself — you should finish in 20 minutes.

1. What is the SI unit of pressure?
2. Who discovered electromagnetic induction and in what year?
3. Escape velocity from Earth is approximately how much?
4. Why does a pressure cooker cook food faster than an open pot?
5. Which type of lens corrects myopia?
6. Speed of sound in air at 0 °C?
7. Which type of mirror is used as a rear-view mirror in vehicles and why?
8. What is the minimum distance needed to hear an echo?
9. State Fleming's Left-Hand Rule (what it applies to).
10. What is 1 horsepower in watts?
11. What radiation has the highest penetrating power — alpha, beta, or gamma?
12. What is the source of the Sun's energy?
13. Who discovered X-rays and in what year?
14. Name three applications of total internal reflection.
15. The sky is blue due to which type of scattering?
16. A transformer works on which principle and which type of current?
17. Which scientist first explained the photoelectric effect?
18. Latent heat of vaporisation of water is approximately how much?
19. What is the commercial unit of electrical energy?
20. Why do stars twinkle but planets do not?
21. Which law of thermodynamics says entropy always increases?
22. Name the person who discovered the neutron.
23. What does SONAR stand for and what does it use?
24. Refractive index of diamond is approximately how much?
25. Indian household electricity supply is at what voltage and frequency?

**Answers:** 1. Pascal (Pa) 2. Michael Faraday, 1831 3. 11.2 km/s 4. Higher pressure raises the boiling point of water to  $\sim 120$  °C; food cooks faster 5. Concave (diverging) lens 6. 331 m/s 7. Convex mirror — because it always gives a virtual, erect, diminished image with a wider field of view 8. 17 metres 9. Applies to an electric motor — Index = Field direction, Middle = Current direction, Thumb = Thrust/motion direction 10. 746 watts 11. Gamma radiation 12. Nuclear fusion (hydrogen nuclei fuse to form helium) 13. Wilhelm Röntgen, 1895 14. Optical fibres, diamonds (sparkle), mirages in desert, periscope 15. Rayleigh scattering 16. Mutual induction; AC (Alternating Current) only 17. Albert Einstein (Nobel Prize 1921) 18. 2260 J/g (at 100 °C) 19. Kilowatt-hour (kWh) — also called 1 unit 20. Stars are point sources — atmospheric turbulence flickers their single ray. Planets are extended discs — their multiple rays average out 21. Second Law of Thermodynamics 22. James Chadwick (1932) 23. Sound Navigation and Ranging; uses ultrasonic waves 24. 2.42 25. 230 V, 50 Hz AC

---

## APPENDIX B — Trap Recognition Card

| Trap                                  | Common wrong answer | Correct answer                                       |
|---------------------------------------|---------------------|------------------------------------------------------|
| First Nobel Prize in Physics          | Einstein, Curie     | Röntgen (1901, for X-rays)                           |
| Nobel winner for photoelectric effect | J.J. Thomson        | Albert Einstein (1921)                               |
| Raman Effect Nobel                    | S.N. Bose           | C.V. Raman (1930)                                    |
| Radio invention — Indian angle        | Marconi alone       | J.C. Bose also demonstrated millimetre waves in 1895 |
| Sun's energy                          | Nuclear fission     | Nuclear fusion                                       |
| Most penetrating radiation            | Beta                | Gamma                                                |
| What stops alpha radiation            | Lead                | Paper or a few cm of air                             |
| Rear-view mirror type                 | Concave             | Convex (wider field)                                 |
| Myopia correction                     | Convex              | Concave lens                                         |
| Speed of sound vs speed of light      | Similar             | Light is $\sim 880,000\times$ faster                 |
| Echo minimum distance                 | 34 m                | 17 m (half the round trip)                           |
| Horsepower in watts                   | 750 W               | 746 W                                                |
| Ice floats because...                 | It is lighter       | Anomalous expansion — water expands on freezing      |
| Geostationary satellite height        | 3600 km             | $\sim 36,000$ km                                     |
| Transformer works on DC?              | Yes                 | No — only AC creates changing flux                   |